

Patchwork Water: Urban Growth, Strained Water Networks, and Household Water Burdens in the Horn of Africa *

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Abstract

This paper describes how urban households secure water when piped networks are unreliable, using a representative sample of 2,197 households in Hargeisa, Somaliland — a rapidly growing, semi-arid city on the frontier of climate stress. Rather than relying on a single source, households assemble “patchwork” portfolios combining piped connections, tanker trucks, rainwater harvesting, and other sources, changing their composition across the dry and rainy seasons. Four findings emerge: (1) source portfolios vary widely across households and seasons; (2) unconnected households pay more and face more shortages; (3) costs and access are largely regressive; and (4) storage is a critical coping asset that poorer households cannot afford. Planned network expansion will narrow, but not eliminate, these inequalities.

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1 Introduction

For decades, expanding piped water access in urban areas has been a central objective of development policy aimed at improving household well-being. Empirical evidence supports this approach: household water connections are associated with improved public health, reduced time spent obtaining water, and enhanced well-being ([Ahuja et al., 2010](#); [Devoto et al., 2012](#); [Galiani et al., 2005](#)). Accordingly, network expansion has remained a cornerstone of urban water policy, with the primary challenge often framed in binary terms: connect unserved households and consider the problem solved.

However, a piped connection does not ensure reliable service. In many rapidly growing cities, achieving universal and dependable piped water is increasingly challenging despite ongoing infrastructure investments. Recent evidence from Latin America shows that connected households often supplement network water with alternative sources, suggesting that conventional access metrics may significantly overstate actual water security ([Martínez-Espíñeira & Pérez-Urdiales, 2025](#)). Similarly, [McKenzie and Ray \(2009\)](#) documented this pattern in urban India two decades ago, where no major city provided continuous piped supply and households invested in tanks and pumps to cope with irregular pressure and quality. Climate change is likely to exacerbate service gaps, as network reliability can deteriorate rapidly during droughts. The *Day Zero* crisis in Cape Town exemplifies this dynamic: as service reliability declined, households increasingly relied on private storage and alternative supplies ([Abajian, Cole, Jack, Meng, & Visser, 2026](#)). These cases reveal a critical limitation: indicators of water security based solely on connection status fail to capture intermittent supply, low pressure, poor water quality, or prolonged service interruptions.

When piped water infrastructure is inadequate, due to either limited coverage or poor reliability, households turn to private solutions and informal sources. Evidence from Nairobi's informal settlements illustrates the coexistence of kiosks, pushcart vendors, tanker trucks, boreholes, legal connections, and illegal connections within the urban water economy ([Crow & Odaba, 2010](#); [Sarkar, 2020](#); [United Nations Development Programme, 2011](#); [Water and Sanitation Program, 2005](#)). In many settings, piped water connections coexist with private wells, and household responses to shortages indicate substantial substitutability between these sources ([Abajian et al., 2026](#); [Pallottini, Wong, & Zhang, 2025](#)).

Reliance on informal water sources can be expensive. Classic work on water vending in Kenya and Nigeria documents that households pay high prices for vendor-supplied water when piped systems are unavailable or unreliable ([Whittington et al., 1991, 1989](#)). Households with sufficient resources may also invest in private solutions such as storage and private wells. Storage enables households to smooth consumption over time to cope with rationing episodes ([Baisa et al., 2010](#); [United Nations Development Programme, 2011](#); [Water and Sanitation Program, 2005](#)). These solutions, however, require significant

upfront investment and are typically inaccessible to low-income households.

Although an expanding literature has shed light on the complexity of urban water security in rapidly growing cities, significant information gaps persist. To our knowledge, no existing studies or surveys have comprehensively documented household water use across the full range of sources, the associated costs, and the heterogeneous patterns across income levels. Large-scale surveys, such as the Demographic and Health Surveys (DHS), consistently record the primary source of drinking water. However, information on secondary sources, non-drinking water sources, prices, and volumes has been inconsistently reported or absent. Research from Nicaragua and other low- and middle-income countries documents investments in storage and alternative supplies among households connected to intermittent networks (Vásquez, 2012; Vásquez & Franceschi, 2013), and broader reviews emphasize the widespread use of vendors, kiosks, boreholes, tanker deliveries, and rainwater harvesting as complementary sources (Majuru et al., 2016). However, none of these studies provide comprehensive documentation of use and expenditures across the *full set* of formal and informal water sources within a city.

A recent study by Martínez-Espiñeira and Pérez-Urdiales (2025) underscores the importance of such documentation. Using data from the Latin American Public Opinion Project (LAPOP) on piped water, bottled water, and tanker truck expenditures, the study shows that conventional measures of affordability and access can be misleading when households rely on multiple, non-exclusive sources. Nevertheless, their analysis primarily focuses on piped and bottled water expenditures, with limited inclusion of tanker trucks due to the small number of users in the 2019 LAPOP wave, further highlighting the need for comprehensive data collection across water sources and within samples that are representative of a city's population.¹ Our study contributes to filling this gap by collecting household-level data specifically designed to characterize a patchwork urban water system, in which both connected and unconnected households combine piped water, kiosks, vendors, tanker trucks, household storage, and rainwater harvesting as part of an integrated portfolio of water sources.

Drawing on a city-representative household survey of over 2,000 respondents in Hargeisa, Somaliland, we document the diverse composition of water portfolios among both connected and unconnected households. Our analysis shows that when network supply is unreliable, households rarely depend on a single source of water. Instead, they assemble portfolios that may include piped connections, tanker trucks, cart vendors, neighborhood sharing arrangements, rainwater harvesting, and private storage. The composition of these portfolios often changes across seasons and in response to variation in prices, availability, and service quality. As a result, securing water becomes an ongoing challenge, requiring households to continuously allocate time, assets, and

¹The sampling frame of this wave structurally excludes anyone without a mobile phone. LAPOP's methodology documentation itself flags that "non-access is generally correlated with sociodemographic characteristics" (Montalvo, Pizzolitto, & Plutowski, 2022).

financial resources across alternative sources.

We refer to this arrangement as a *patchwork water system*: a decentralized combination of public, formal, and informal providers that emerges as households adapt to supply constraints. Rather than replacing the public network, alternative sources often complement it. Households use them to reduce exposure to supply interruptions, smooth seasonal fluctuations, and manage uncertainty about future water availability. Understanding how these portfolios are assembled, what their cost implications are, and how they affect household vulnerability to shortages is central to understanding household water resilience in growing urban populations.

Hargeisa provides a valuable setting for studying these dynamics. Over the past two decades, the city has undergone rapid population growth while facing severe water constraints due to its arid climate and limited local water resources.² The rapid urbanization pattern is typical of many other medium-sized cities in Africa. In response to this growth, the Hargeisa Water Agency (HWA) has invested substantially in network expansion but also adopted a permanent water schedule whereby households receive piped water on a fixed number of days per week. These water restrictions will apply even to households that previously received water daily. While the public network will remain a key component of the city's water system, many households will continue to depend on tanker deliveries, cart vendors, private storage, and rainwater harvesting.

Our analysis of survey data and qualitative fieldwork observations leads to four main takeaways. First, we document that most households—including many that are connected to the network—rely on multiple water sources. The portfolio of sources changes drastically across seasons, with rainwater featuring prominently during the rainy season as the most prevalent secondary water source. In semi-arid Hargeisa with an annual average precipitation of 388 mm, a surprising 60% of households in our survey report relying on rainwater. In the dry season, the large majority of households rely on a single source.

Second, although a network connection does not guarantee a consistent water supply due to intermittency, connected households—especially those with more frequent service—experience fewer shortages, lower overall water costs, and improved access. Third, we observe significant regressivity in both the cost of and access to water, attributable in part to the disproportionate benefits of network connections for high-income households and to source-specific pricing.

Fourth, we identify storage as an important asset that enables households to manage intermittency across all water sources and maximize rainwater harvesting during the rainy season. Storage capacity increases with lack of network connectivity, except for low-income households. This pattern suggests that income is a significant constraint on resilience in this context.

²Hargeisa is located in a semi-arid region where rainfall is low, highly seasonal, and characterized by substantial inter-annual variability.

These takeaways contribute to our understanding of water resilience in a rapidly growing city characterized by limited network connections and low reliability—features common to many cities in the developing world. Drawing on our results, we consider the implications of current plans to expand the network to 60% of households while simultaneously reducing supply reliability for all connected households. The planned expansion is likely to reduce some of the regressivity observed in the survey by extending lower-cost piped access to areas with a more balanced income composition and higher current shortage rates. However, these improvements will be partial: households outside the planned service boundaries, who are disproportionately low-income, will remain excluded, and reduced service frequency for currently connected households may sustain the need for costly backup sources and private storage. Expanding access in a way that does not perpetuate existing inequities may require considering the full portfolio of choices households face when network reliability is insufficient.

2 Data and Methods

The objective of this paper is to describe residential water supply and demand in Hargeisa, Somaliland, and how it varies across seasons, levels of network connectivity, and income levels. Our descriptive analysis draws from two main sources. A household survey and complementary fieldwork observations that accompanied the survey, including visits and interviews with private water providers.

2.1 Hargeisa water survey

We conducted a survey across 2,197 households in November and December of 2025. In order to obtain a comprehensive picture of Hargeisa’s water sources at the household level across income and network connectivity, our sample was stratified based on income levels and network water connections at the zone level.³ All our analyses use sampling weights to generate population-representative statistics.

The survey contains five modules. Module 1 asks about basic household characteristics, including household size, income level, education level, occupation, house ownership, and toilet facilities. Module 2 concerns water sources and costs of obtaining water. It documents the primary and secondary water sources each household relies on in the dry and rainy seasons, respectively. It also asks the monetary and time costs of obtaining water from various sources. Module 3 specifically targets rainwater harvesting, collecting information on harvesting behavioral patterns, the infrastructure

³Consistent with planned stratification, we found significant differences in average income levels and connectivity status across strata. However, we also found substantial overlap of the income and connectivity distributions across strata. Thus, rather than using zone-level income and connectivity to examine heterogeneity in the variables of interest, we use income and network connectivity *at the household level*.

for rainwater capture, storage, and treatment, and willingness to pay for an upgraded complete rainwater harvesting system. Module 4 concerns water shortage experiences, asking about the frequency with which households experienced shortages for an array of needs and the negative impacts on work, schooling, and mental distress. Lastly, Module 5 asks for household service hiring and attitudes toward vocational training.

We supplemented our survey with a phone follow-up of 294 households to further inquire about water prices and storage. They were randomly selected from the main survey sample, following a stratified sampling scheme by household-level income and water connection. The home-to-home main survey described above asked households that reported having collected rainwater for both movable and permanent storage capacity, but collected only mobile storage for those that reported not using rainwater as a source (about 60% of the sample). However, upon analyzing the data, we realized that both storage types are important components of all households' water portfolios regardless of sources used. Hence, we asked for the storage capacity of both types via a telephone re-survey of 99 households that reported not using rainwater as a source in the main survey. We also found that the water cart price data collected in the original survey was not consistent with the data we collected in the field from asking water cart providers directly. We therefore re-surveyed 195 households by phone, asking for cart and tanker truck prices with explicit attention to units of measurement. All figures and tables that use this supplementary survey adjust survey weights accordingly based on our sampling strategy for this subsample of participants.

2.2 Qualitative surveys and observations

During May 2026, we conducted an in-field investigation in Hargeisa, Somaliland, to improve our qualitative understanding of the household water market and collect pricing information for the supply side of the market. Our team held conversations with a number of water cart providers and tanker truck drivers to collect their fixed costs, operational costs, and revenue. We also visited upstream locations where these water retailers obtain water, such as water kiosks and privately-owned boreholes. Interviews with officials at the HWA, the authority regulating the network, provided detailed knowledge of its expansion history and plans. We also collected prices of water storage tanks of different capacities from two leading tank manufacturers and retailers.

3 Four Takeaways on Urban Water Insecurity in Hargeisa, Somaliland

We organize the analysis of the survey and qualitative observations around four main takeaways. First, we find that households rely on a patchwork of water sources with wide variation in source use across households and seasons, with rainwater playing a surprisingly large role. To our knowledge, this has not been documented in urban settings. Second, we find that households with no connection, or with a connection but low reliability, pay more for water and suffer more water shortages. Third, we find that—partly because high-quality and low-quality water connections are more common in the upper segment of the income distribution—the heterogeneity in water prices and access translates into regressivity by income. Fourth, we find that—because nearly all sources are intermittent—storage plays a very large role in securing water access. Storage increases with the absence or unreliability of network connections. However, low-income households own low volumes of storage, which limits their ability to collect rainwater in the rainy season and arbitrage across water sources in the dry season.

3.1 Takeaway 1: Most households rely on a patchwork of sources that vary across households and across seasons

Water consumption in Hargeisa makes use of an intricate "patchwork" system: households rely on a heterogeneous mix of water sources, combine them in complementary ways, and alter those combinations across the dry and rainy seasons. The portfolio of sources shrinks during the dry season, when rainwater—a dominant secondary source—is no longer available.

A variety of sources

Figure 1 describes the distribution of primary (horizontal axis labels) and secondary (vertical axis scale) water sources across seasons. Five leading water sources emerge in Hargeisa: private tanker trucks, piped water connections at the household level, donkey- or motorcycle-powered carts, neighbor sharing, and rainwater. Notably, as we describe in this section, all of these sources are intermittent (i.e., available to households less than 100% of the time). In contrast with other settings, such as India or South Africa ([Abajian et al., 2026](#); [Sekhri, 2014](#); [Subramanyam, 2024](#)), private wells are completely absent from the water portfolio even among the wealthiest homes. One possible reason for this is that it is difficult to find groundwater in Somaliland that is under the WHO-recommended salinity threshold ([McSorley & Zakaria, 2022](#)).

Tanker trucks constitute the most important primary source of water, with about 65%-70% of households relying on them as their primary source. Tanker truck owners

purchase water from a number of private water boreholes 40-50 km to the east and the west of Hargeisa, and act as intermediaries between private underground sources and households, charging a markup of about 700%-800%. A tanker truck usually hauls a water tank of 9-12 m^3 and delivers water to tanks at households in Hargeisa, who can order water via phone call or text message. The market of private boreholes and tanker trucks is still relatively unregulated: HWA has attempted to impose chlorine disinfection requirements for water in incoming trucks, but has faced obstacles with enforcement. A tanker truck association exists, but its functionality is rather limited according to several drivers interviewed.

Roughly 22% of households rely on piped water as their primary source, extracted from 16 boreholes in wellfield areas 23-25 km to the north of Hargeisa, currently covering about 35% of Hargeisa's population.⁴ Most of the current connections are concentrated in the older parts of the city (see **Figure 6**), with pipelines constructed in the 1970s under an economic cooperation agreement between China and the Somali government. In addition, water supply was intermittent, with 71.5% of connected households in our survey receiving piped water fewer than 3 days per week. Water extraction, transmission, household-level connection, and pricing all fall under the purview of HWA.

Water from water carts constitutes a relatively small share of the water market, with around 4% of households reporting it as their primary source. Water cart owners typically obtain water from privately owned water stands (also called water kiosks) in network-connected neighborhoods. Water stands have a piped water connection and purchase water from HWA. Kiosk owners then sell water to cart vendors at a markup of roughly 80% - 130%. Donkeys, motorcycles, and small pickup trucks are common power sources for cart vendors, which typically tow water tanks of 0.2-1.4 m^3 capacity. Cart vendors then sell at a markup of 150%-250% to households making orders by telephone calls.

Rainwater - which is seasonal - is only used as a primary source during the rainy season by about 4.6% of the households. However, as we discuss next, it plays an important role as a secondary source when rainfall is available. This availability is concentrated in Hargeisa's two rainy seasons: the main rainy season, *Gu*, which takes place between April and June, and the shorter *Dayr* season, between October and November. Average annual precipitation is estimated at 388 mm. Rainwater is collected largely through makeshift rainwater collection systems whose characteristics vary widely across households. These systems range from collecting through open containers at ground level to partial roof collection using a funnel and a small movable tank to very sophisticated systems with underground storage and filtering mechanisms. Finally, the majority of households own some water storage (see Subsection 3.4), which allows them to share

⁴A series of expansionary and upgrading projects of the piped water network have been completed, with one remaining project reaching the final stage. Refer to Section 4 for more details on the completed and projected expansion phases, and their implications on household water consumption and expenditure.

stored water with neighbors.

Source-stacking and seasonality

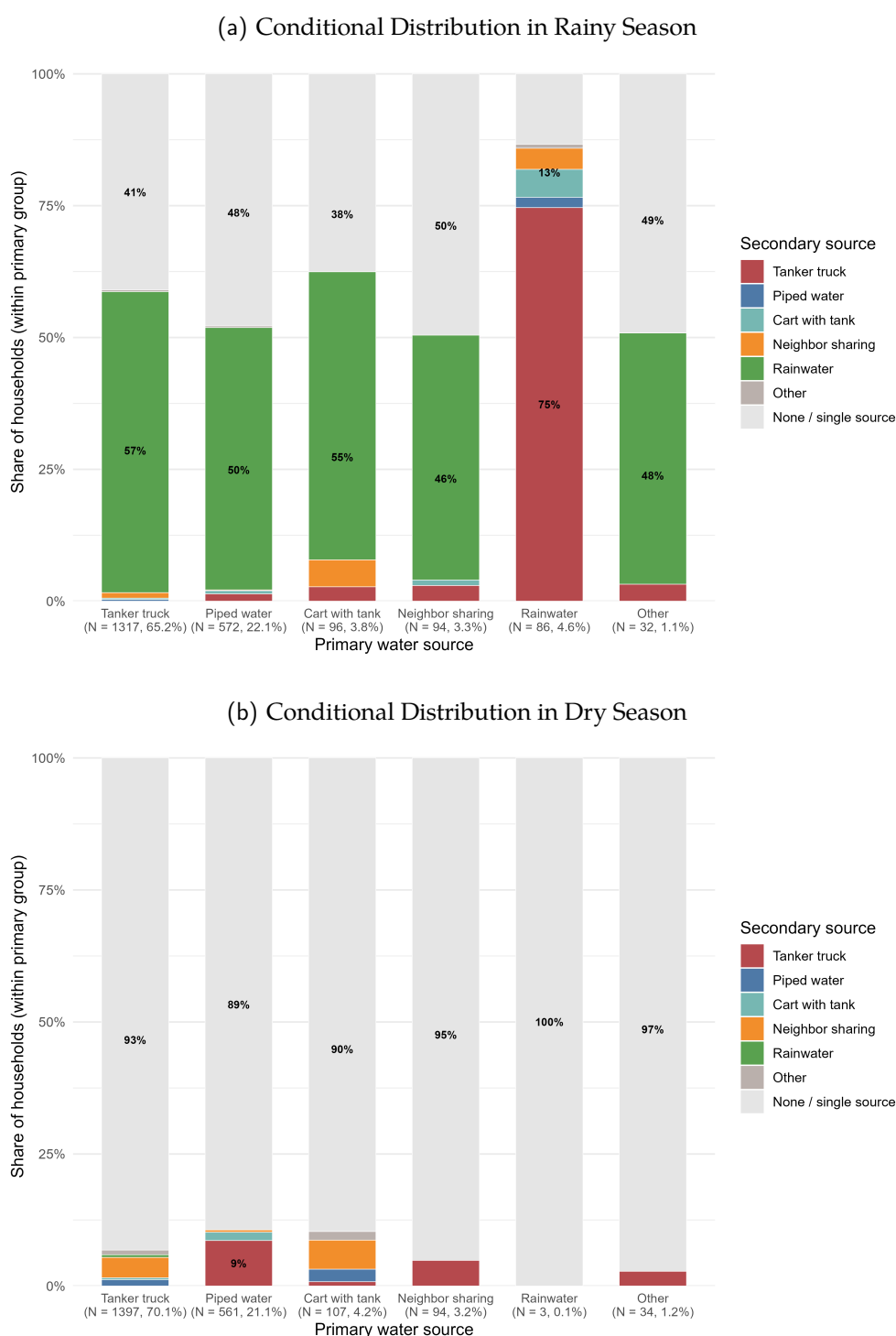
Households do not rely on a single water source. Most combine multiple sources, and the mix shifts sharply across seasons. During the rainy season, the vast majority of households stack a primary source with rainwater; during the dry season over 90% depend on a single source. Figure 1 reveals two key patterns. First, across all primary source types, the majority of households report a secondary source during the rainy season (**Figure 1 (a)**), whereas over 90% of households use a single source in the dry season **Figure 1 (b)**. Second, the composition of secondary sources is heavily dominated by rainwater: 46–57% of households whose primary source is piped water, tanker truck, cart with tank, or neighbor sharing report rainwater as their rainy-season secondary.⁵ When rainwater is unavailable during the dry season, tanker trucks and neighbor-sharing become the primary backup sources.

The dry season exacerbates water stress in multiple ways. Tanker truck owners report that waiting times at boreholes spike to 2 hours during the dry season compared to 1 hour during the rainy season, as pumping times slow down. These delays translate into sparser supply or higher prices for households. In addition, per-liter costs vary dramatically across sources, as discussed in Subsection 3.2, with rainwater priced between piped water and tanker truck rates. These facts point towards increased fragility of water access during the dry season.

The patchwork structure emerges directly from this seasonality: households combine multiple sources to smooth supply fluctuations across the year, with rainwater availability driving the seasonal composition of their water portfolios.

⁵Among the small group of rainwater-primary households, 75% report tanker truck as their secondary source.

Figure 1: Conditional Distribution of Secondary Water Sources



Note: Each bar shows the population-weighted distribution of secondary water sources among households reporting the indicated primary source, separately for the indicated season. Bar labels on the horizontal axis show the unweighted number of households (N) and the weighted share of the population reporting each primary source. Source categories collapse "Public tap/standpipe/kiosk" and "Natural source" into "Other"; "None / single source" indicates households reporting no secondary source. Shares are computed using inverse-probability weights calibrated to stratum population sizes.

3.2 Takeaway 2: Unconnected households pay more for water and suffer more from water shortages.

Although piped water connections are imperfect, as is evident by the heavy reliance on secondary sources by households whose main water source is piped water, expanding network connections can potentially increase water security and lower prices per liter. Our data show that piped water connections are associated with fewer self-reported instances of water shortages and lower prices per liter of water consumption excluding rainwater.

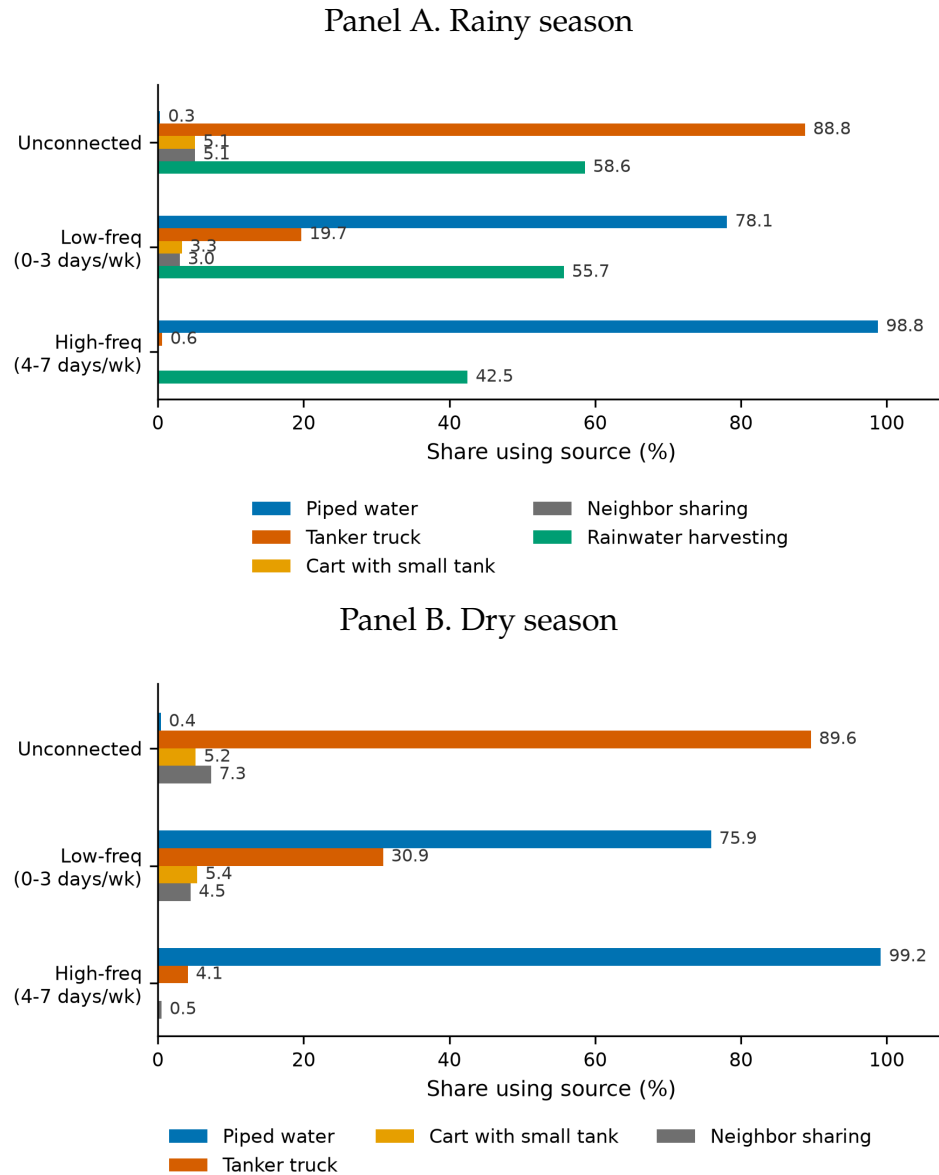
Degrees of connectivity

Being "connected" in Hargeisa rarely means receiving continuous piped service. Of the 652 households reporting a piped connection, only 186 (29%) receive water 4–7 days per week; the remaining 466 receive water 0–3 days per week. Reliable piped service is therefore the exception even among the connected minority, and this intermittency is itself a likely structural driver of the multi-source patterns documented below — when piped water arrives only a few days a week, households must look elsewhere to meet daily needs.

Two patterns emerge from **Figure 2** that deepen the patchwork story. First, comparing unconnected and connected households reveals a rough substitution between tanker truck delivery and piped water. The 1,545 unconnected households — the majority of the sample — rely overwhelmingly on tanker truck delivery (89.6% in dry season, 88.8% in rainy season), while only about 0.4% report piped water as either primary or secondary. Among the 652 connected households, the dependence on tanker delivery falls sharply (to 0.6–30.9% across frequency groups and seasons), with piped water taking its place. Connection to the piped network thus appears to substitute for, rather than complement, tanker truck delivery.

Second, even within the connected population, multi-source reliance persists. Among low-frequency connection households (0–3 days per week), 19.7–30.9% still use tanker delivery as a primary or secondary source, 3.3–5.4% use carts, and over half (55.7%) supplement with rainwater in the rainy season. Even high-frequency connection households (4–7 days per week) supplement with rainwater (42.5% in the rainy season) and occasionally use tanker delivery (0.6–4.1%). The patchwork therefore does not disappear with piped access; it scales inversely with piped service reliability.

Figure 2: Primary and secondary water source and degrees of connectivity by season



Note: Households are grouped by self-reported piped-water connection reliability: unconnected ($N = 1,545$), low-frequency service (0-3 days/week) ($N = 466$), and high-frequency service (4-7 days/week) ($N = 186$). Values labeled at the right of bars are population-weighted shares of households among each connectivity group reporting each source as a primary *or* secondary water source in the indicated season.

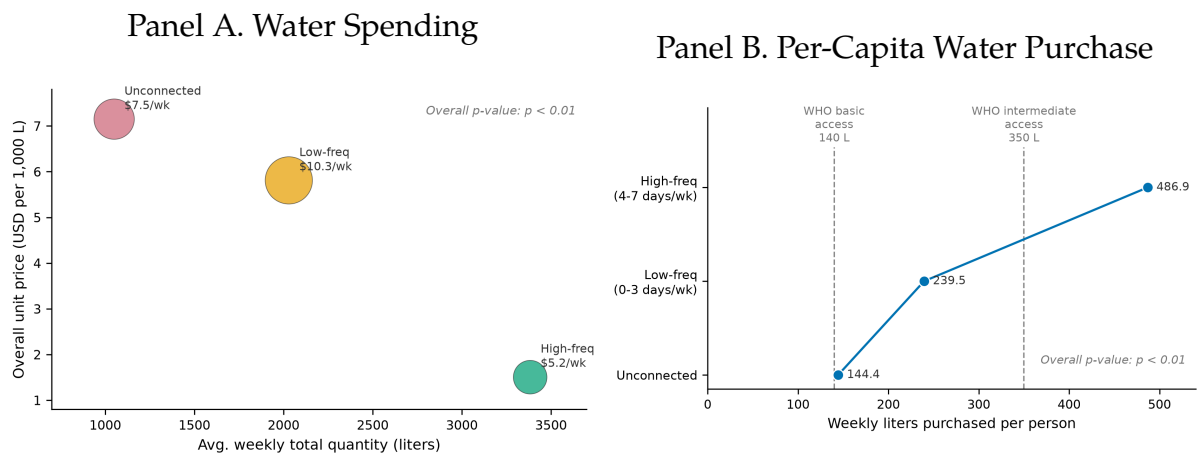
The cost of no connection and infrequent service

We next examine how connection status affects household water purchases, expenditure, and prices. Our survey collected information on the price and expenditure associated with households' most recent purchases from tanker trucks, water carts, and piped water—the three primary water sources discussed in Takeaway 1. Combined with reported purchase frequency, these data allow us to estimate weekly water purchases. **Figure 3** Panel A reports average weekly purchased quantity, average weekly expendi-

ture, and the average overall unit price across households with different levels of water connectivity.⁶

Households without reliable water connections purchase less water and pay substantially higher prices. First, unconnected households purchase only 31% as much water per week as reliably connected households, while households with unreliable connections purchase about 60% as much. Second, unconnected and unreliably connected households face substantially higher unit prices, paying 373% and 284% more per 1,000 liters, respectively, than reliably connected households. Third, Panel B from **Figure 3** displays average per-capita water purchase by connectivity. For unconnected and unreliably connected households, per-capita water purchase is only 144 and 240 liters per person per week, respectively, just above the WHO Basic Access threshold of 140 L/person/week (Howard et al., 2020). By contrast, reliably connected households average 487 L/person/week, exceeding the WHO Intermediate Access threshold of 350 L/person/week. Taken together, households that are unconnected or have unreliable service spend significantly more on water and consume half or less water compared to households with reliable connections.⁷

Figure 3: Water Cost by Degree of Connectivity



Note: Panel A plots, for each connectivity group, average weekly total quantity of water purchased (x-axis) against the average overall unit price (y-axis); dollar labels at the points report average weekly water spending. Panel B reports average weekly liters purchased per person for each connectivity group. Total water purchase sums water bought from cart, tanker, and piped utility. Utility quantity is derived using the Hargeisa Water Agency piped-water tariff. All values are population-weighted. P-values are design-adjusted Wald tests of the joint null that all connectivity-group dummy coefficients equal zero in a survey-weighted linear regression.

⁶Panels D, E and F of Appendix **Table 1** show the same statistics disaggregated by source. Note that the amount of water consumed may differ from the amount of water purchased (reported in **Figure 3** and Panel G of Appendix **Table 1**) due to rainwater harvested. Although we did not collect information on the amount of rainwater harvested, storage (which we discuss in Section 3.4) largely determines how much rainwater is collected.

⁷Detailed comparisons across connectivity levels in household characteristics, storage capacity, by-source water purchase are in the Appendix **Table 1**.

No network water results in shortages

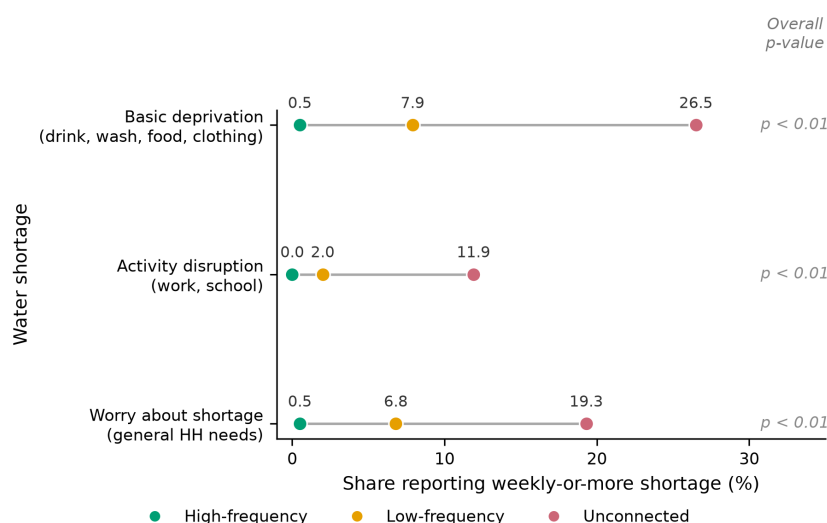
Although **Figure 3** suggests that even unconnected households have an average weekly consumption that is above WHO's basic access, it masks recurrent shortages produced by the intermittency of water sources in Hargeisa. To examine how network connectivity relates to household water security, we measure the frequency of water shortages by asking households whether they experienced a series of water-related hardships during the previous four weeks. These experiences fall into three broad categories. First, basic deprivation, including lacking sufficient water for drinking, hand and body washing, food preparation and cooking, or clothes washing. Second, activity disruption, including missing work or school because of water shortages. Third, concerns about water shortage, measured by how often household members were worried about having insufficient water for their general household needs.

Figure 4 reports the distribution of households by reported shortage frequency across the three connectivity groups.⁸ Water insecurity increases sharply as network connectivity deteriorates. Across all three dimensions, households without a reliable network connection are substantially more likely to report recurrent water shortages. For example, 26.5% of unconnected households report weekly or more frequent episodes of basic deprivation, compared with only 0.5% of reliably connected households. Similar gradients are observed for activity disruption and worries about household water shortages.

In sum, our survey data suggests that although network connections are scarce and imperfect, they constitute a clear way of reducing the water expenditure burden and providing additional water security for households. Connecting additional households to the network remains a central goal of HWA. Several phases of network expansion and upgrading have been completed or are underway. We discuss the scope of these expansions in [Section 4](#).

⁸See also Panel C of Appendix **Table 1**

Figure 4: Shortages by Degree of Connectivity



Note: Values are population-weighted shares of households reporting weekly-or-more water shortages during the previous four weeks. Basic deprivation includes insufficient water for drinking, body washing, food preparation, or clothes washing. Activity disruption includes missing work or school because of water shortages. Worry refers to concern about insufficient water for general household needs. P-values are design-adjusted Wald tests of the joint null that all connectivity-group dummy coefficients equal zero in a survey-weighted linear regression.

3.3 Takeaway 3: The patchwork system is largely regressive

Given that access to piped water shapes the costs, volume and reliability of water supply, those groups that disproportionately benefit from connections will have better access to water. In this section we show that high quality network connections are more prevalent among high income households, and that this regressivity trickles down to all other dimensions of water access as unconnected households are forced to use more expensive and less reliable sources. Moreover, we document an additional source of regressivity for low-income households: source-specific prices.

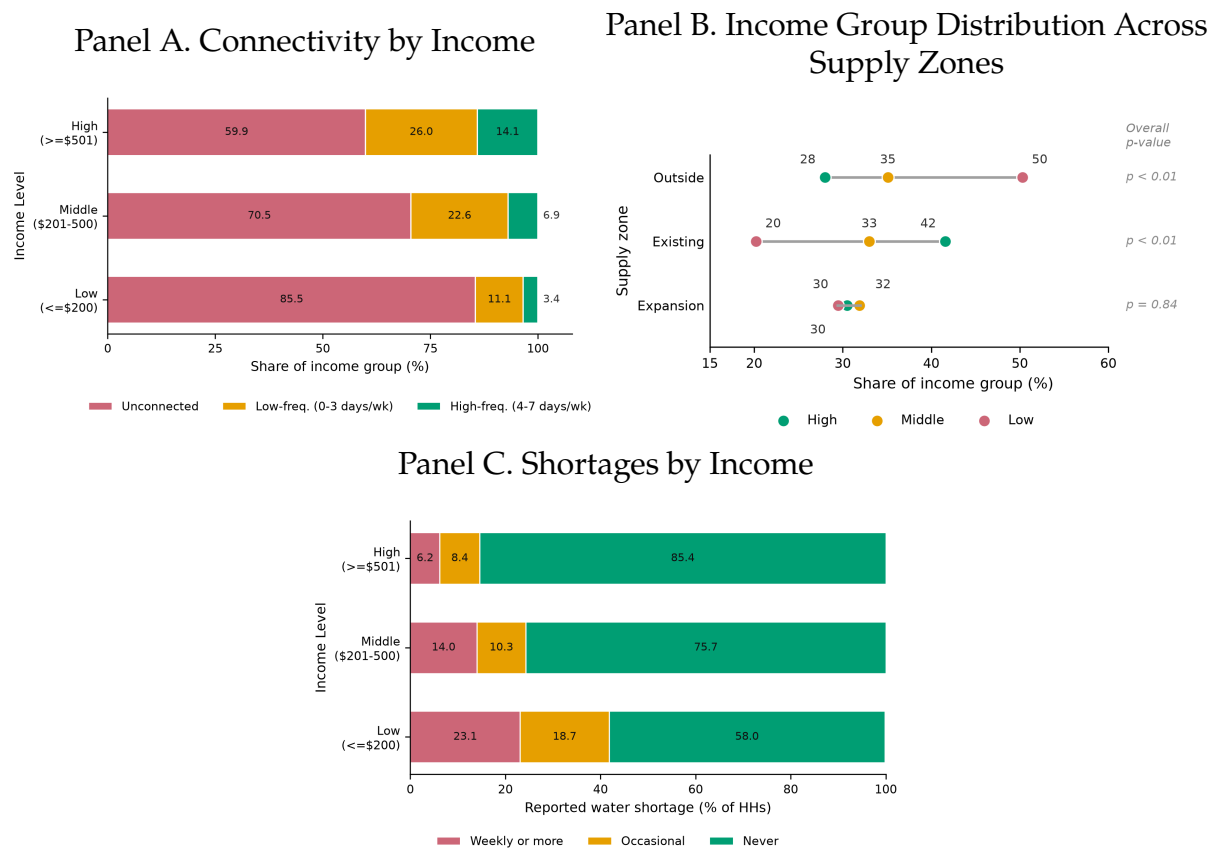
Access to piped water is regressive, both in terms of connections and reliability

Figure 5 shows water connectivity and reliability by income.⁹ First, piped water access is sharply regressive on both the extensive and intensive margins as suggested by Panel A. Low-income households are substantially more likely to be unconnected (85,5% vs. 70.5% vs. 59.9% for low, middle, and high income, respectively). Among the connected, low-income households are also disproportionately on the low-frequency side of the service spectrum (11.1% low-frequency vs. 3.4% high-frequency for low-income, compared with 26% vs. 14.1% for high-income). The high-frequency reliable connection — the closest thing to a continuously available piped supply — is about

⁹See also Appendix **Table 2**

three times more common in the highest-income group than in the lowest.

Figure 5: Regressivity of the patchwork water system



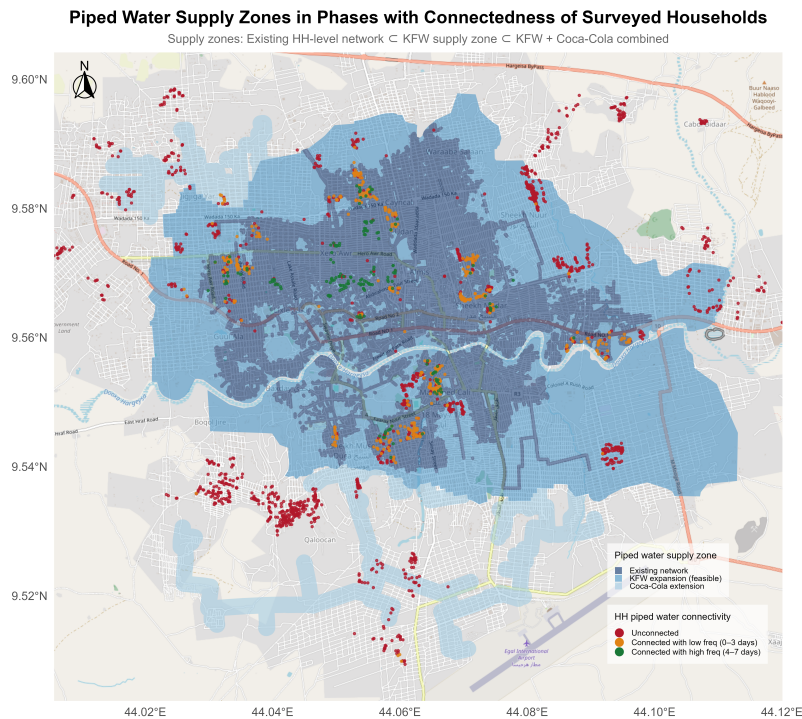
Note: Values are population-weighted shares by self-reported monthly income group. Low income is $\leq \$200$ ($N = 789$), middle income is $\$201\text{--}\500 ($N = 1,009$), and high income is $\geq \$501$ ($N = 393$). Panel A reports household-level piped-water connection reliability. Panel B reports each household's location relative to existing, planned expansion, and outside supply zones. Panel C reports shortage frequency for general household water needs during the previous four weeks. P-values are design-adjusted Wald tests of equality across income groups from survey-weighted regressions.

Second, the uneven distribution of connections across income groups is related to the shape of recent urban growth, with lower-income households concentrated outside the range of the existing water network. We divide Hargeisa into three zones as displayed in **Figure 6**: existing supply zone of piped water; expansionary supply zone of piped water to be completed by the end of 2026, which is not yet operational when the survey was conducted; and outside region unserviceable according to current plan. Panel B from **Figure 5** shows that low-income households are disproportionately located outside the piped infrastructure footprint (50.3% in the Outside zone, compared with 28% of high-income households), while affluent households cluster in the existing network area (41.6% vs. 20.2%). The planned expansionary area shows a roughly balanced

income composition.¹⁰

Third, Panel C from **Figure 5** suggests that water shortages track income in the same regressive direction as piped water access. Frequent shortage (weekly or more) is reported by 23.1% of low-income households against 6.2% of high-income households, a 3.7-fold gap. The regressiveness of piped water access in Hargeisa therefore operates through multiple, mutually reinforcing channels: poorer households are less likely to be connected, live in less-served areas, and face more frequent shortages.

Figure 6: Piped Water Supply Zones and Household-level Connectedness



Note: The map classifies household locations by piped-water supply zone: existing network area, planned expansion area, and outside the current network expansion plan. Household points indicate survey respondents' reported piped-water connection status and service frequency.

Low-income households pay more per liter, and consume less water

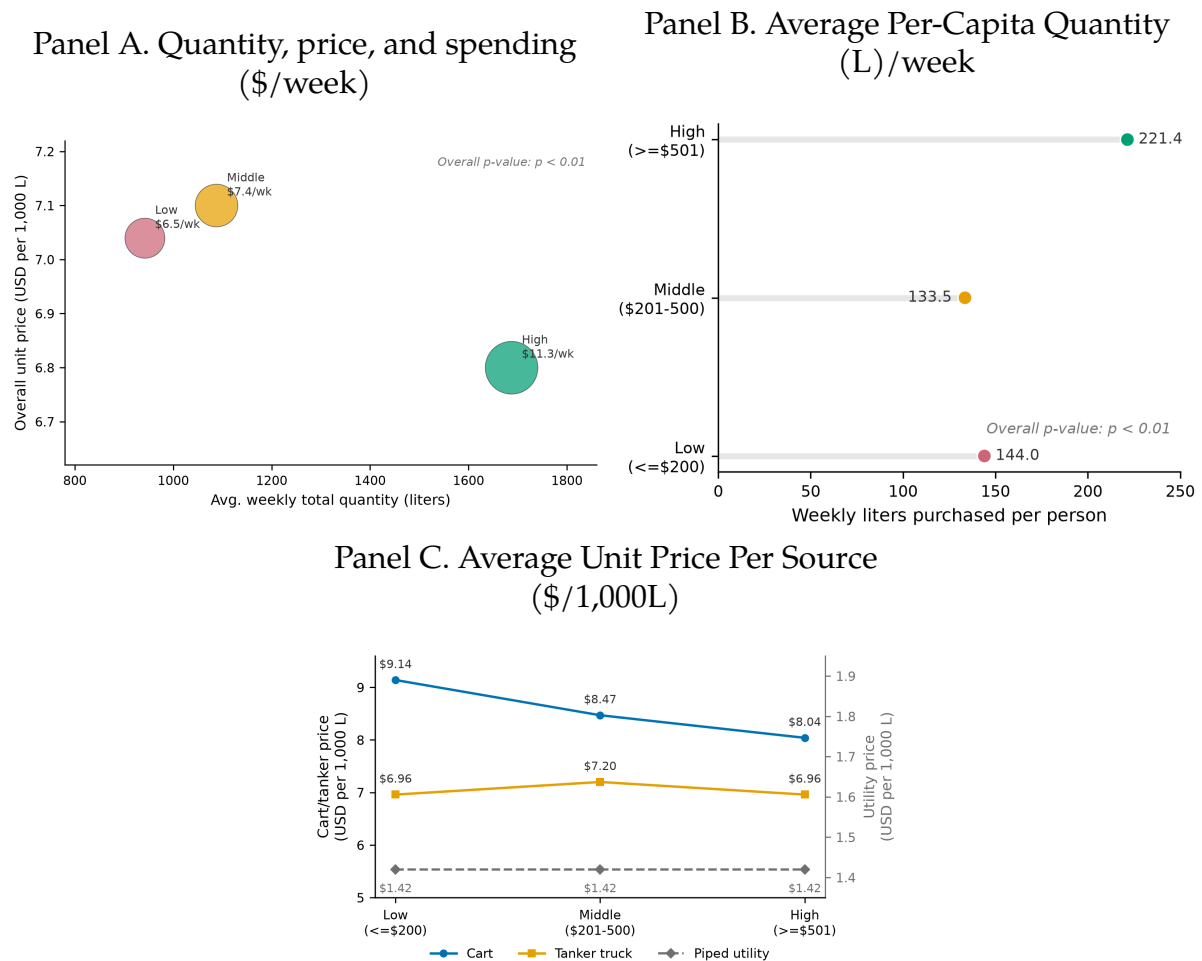
We now explore whether prices per liter are disproportionately higher for low-income households, and how this affects their consumption. **Figure 7** shows weekly water spending, quantities, and unit prices by income group. Panel A shows that, on average, low-income households purchase substantially less water per week than high-income households.¹¹ In addition to purchasing lower quantities, they also face a higher overall unit price per 1,000 L. Panel B shows that average per-capita purchased quantity is about 1.6 times higher for high-income households than for low- and middle-income

¹⁰We introduce network expansion projects in Hargeisa in Section 4.1, and discuss the consequences of the network expansion in Section 4.2.

¹¹See also Panel G in Appendix Table 2

households. Panel C suggests that households face different water prices depending on their income level. This is particularly true for water from cart vendors. Although tanker truck prices are broadly similar across income groups, and the piped water tariff is flat, these prices represent a higher burden as a share of household income for low- and middle-income households, which is consistent with their lower purchased quantities.¹²

Figure 7: Regressiveness of quantities and prices by income



Note: Values are population-weighted and reported by self-reported monthly income group. Low income is $\leq \$200$ ($N = 789$), middle income is $\$201-\500 ($N = 1,009$), and high income is $\geq \$501$ ($N = 393$). Panels A and B use total weekly water purchase measures among resurveyed households, summing purchases from cart vendors, tanker trucks, and piped utility water, with non-purchase from a source coded as zero. Panel A plots average weekly total quantity of water purchase (x-axis) against the average overall unit price (y-axis); dollar labels at points report average weekly water spending. Panel B reports average weekly liters purchased per person. Panel C reports source-specific unit prices conditional on being an observed user or payer of each source. Cart and tanker prices are restricted to resurveyed users of each source. Piped utility quantity is derived from reported utility spending using the Hargeisa Water Agency tariff at $\$1.425/m^3$ as shown in Panel C.

¹²Detailed breakdown of water purchase by water sources can be found in Panels D, E and F of Appendix Table 2.

These patterns show that two related channels contribute to the regressiveness of water access. First, households without a connection, or with less reliable service, rely more heavily on non-piped sources, which are often more expensive than network water. Second, for some of these sources, low-income households face higher unit prices than higher-income households. These higher prices constrain the quantity of water households can afford to consume, leaving poorer households with a higher expenditure burden and greater exposure to shortages.

3.4 Takeaway 4: Large volume storage constitutes an important coping mechanism, but is less prevalent among the poor

Takeaway 1 shows that the patchwork system Hargeisa relies on a variety of intermittent sources that complement existing network connections and substitute for absent connections, while takeaways 2 and 3 show that some of these sources are more expensive than piped water, resulting in disproportionate burdens for unconnected and low-income households. The intermittency of all sources, including rainfall, which comes at a marginal cost of zero, means that shortages are unavoidable in the absence of on-site storage. This section discusses how households have adopted storage as a coping mechanism in response to limited and infrequent network connections, and how income constitutes a barrier for adoption.

The many versions of storage

During fieldwork, we observed substantial variation in the types of water storage used by households in Hargeisa. The most common storage devices include jerricans, barrels, and large-capacity water tanks, which differ considerably in storage capacity, functionality, and acquisition cost.

Jerricans are 20-liter plastic containers with flat sides and built-in handles that facilitate both small-scale water transport and storage. Barrels are plastic drums with a storage capacity of approximately 200 liters. They may or may not be fitted with lids depending on household conditions and typically do not include faucets. Owing to their relatively low purchase cost, jerricans and barrels are the predominant storage technologies among low-income households.

Large-capacity storage tanks are typically constructed from plastic or metal and range from 1,000 to 2,500 liters in capacity. Unlike smaller containers, these tanks are generally equipped with fitted lids and outlet faucets, making them more suitable for long-term household storage. However, they require substantial upfront investment, with retail prices ranging from approximately US\$150 to US\$450 depending on the material, storage capacity, and design. These high fixed costs can limit households' ability to expand storage capacity, particularly among low-income families. At the upper end of the storage spectrum, some households have permanent storage infrastructure, including

underground cisterns (*berkads*) and cement or ferrocement tanks. These systems require specialized construction and substantially higher upfront investment than portable storage containers, making them an option primarily for better-off households. In the following analysis, we report total storage capacity aggregated from all types.

Storage, network connectivity, and income

Figure 8 displays the overlap between average total storage capacity, piped-water connectivity, and income level. The figure highlights an important interaction between need and ability to cope. Among households with the greatest need for storage - those who are unconnected or receive low-frequency piped service - storage capacity largely increases with income. Unconnected high-income households hold about 2.2 times as much storage as unconnected low-income households, while among households with low-frequency service the high-to-low income ratio is about 2.4.

This pattern suggests that storage is an important coping asset, but one that is less available to poorer households precisely where it is most valuable. Low-income households without reliable piped service have substantially smaller storage capacity than middle- and high-income households facing similar connectivity conditions. This likely exacerbates the inequalities documented above: poorer households are more exposed to unreliable water access and have less capacity to smooth that unreliability through storage.

The relationship looks different among high-frequency connected households. For households with reliable piped service, the need to invest in large storage capacity is lower. Consistent with this, high-income households with high-frequency service hold far less storage than high-income households who are unconnected or receive low-frequency service. More broadly, the figure suggests that storage reflects both exposure to unreliable service and the resources available to cope with that exposure.

Storage is a year-long coping mechanism

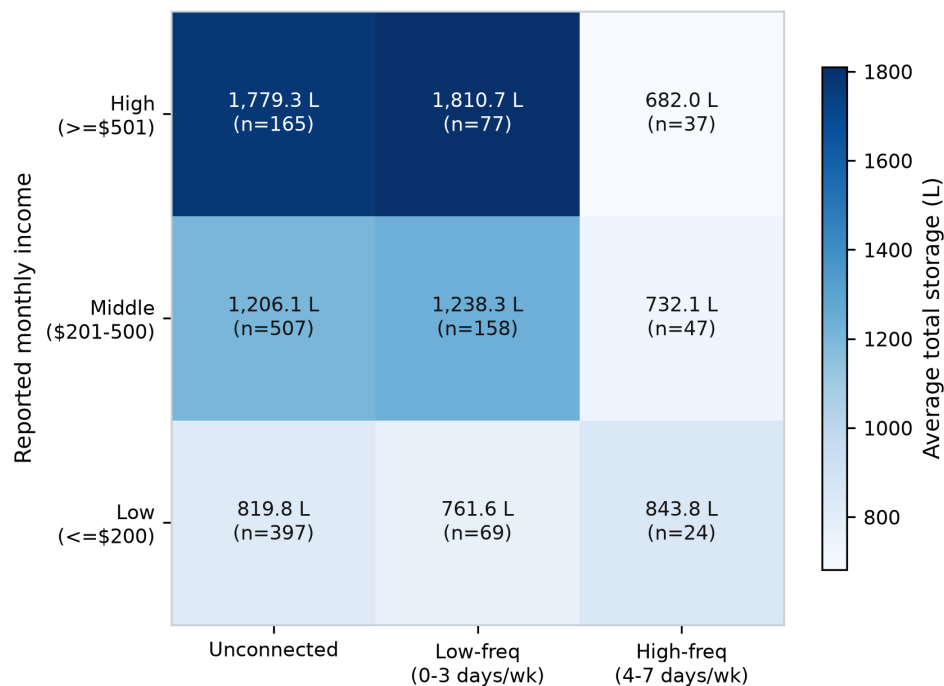
Storage is a valuable asset during both rainy and dry seasons. First, unconnected households largely depend on tanker trucks (about 89%) and cart vendors (5.2%), which requires households to have storage room for purchased water, particularly during dry months. Similarly, during the rainy season, all income households, and particularly those unconnected and those with low-frequency service, complement their water portfolio with rainwater. Rainwater harvesting capacity is similarly constrained by storage. This points out the critical role of storage throughout the year as it allows households to store rainwater when available and to smooth purchases from tanker trucks, carts, and neighbors when piped water is absent or unreliable.

Storage may also matter for the prices households face. Low-income cart users pay higher unit prices (14%) than high-income cart users. One possible reason for this is

that households with less storage have less flexibility over purchase timing and quantity, making them more dependent on small or urgent purchases, which can limit their ability to negotiate prices.

Taken together, the evidence points to storage as a coping asset that interacts with both piped water infrastructure access and household resources. Poorer households are more likely to face unreliable water access, rely on expensive non-network sources, and hold smaller storage capacity. Thus, large volume storage does not simply reflect household wealth: it is also part of the technology that households use to manage water scarcity. Unequal access to this technology can amplify the burdens of weak network connectivity.

Figure 8: Storage capacity (L) by connectivity and income levels



Note: Cell values report population-weighted average total storage capacity in liters. Parentheses report household counts in each income-connectivity cell. Computation of storage means includes households with zero storage, i.e., unconditional on positive storage.

4 An eye towards the future

The future of Hargeisa's urban water system will depend largely on the continued expansion of the piped water network. This section provides an overview of completed and ongoing network expansion projects and discusses their potential implications for household water access and consumption in the coming years.

4.1 Overview of network expansion projects

The original piped water network in Hargeisa was constructed between 1972 and 1974 under an economic cooperation agreement between the Chinese and Somali governments. The system conveys groundwater from the Geed Deeble wellfields through two parallel 300 mm (12-inch) transmission pipelines spanning approximately 23–25 km to a large storage reservoir located in the northwestern part of Hargeisa, commonly known as the Chinese Reservoir. Based on our survey, 22% of households are connected to this original network (shown in dark navy color in **Figure 6**). However, the system faces several challenges, including limited transmission capacity, aging infrastructure, and deterioration of water quality associated with obsolete pipe materials. In response, HWA, together with multiple international development partners, has undertaken a series of major investments to upgrade water extraction and transmission, expand neighborhood coverage, and increase household connection through three flagship projects.

Hargeisa Urban Water Supply Upgrading Project (HUWSUP). Implemented between September 2015 and October 2022, HUWSUP focused primarily on increasing bulk water extraction and transmission capacity. Its major components included: (1) construction of 16 new boreholes in the Geed Deeble, Hora Haadley, and Laas Duur wellfields, located 23–25 km north of Hargeisa; (2) installation of a new 600 mm transmission pipeline connecting the wellfields to the Chinese Reservoir, replacing the old transmission lines built in the 1970s; and (3) construction of modern pumping stations to support high-volume water conveyance.¹³

Coca-Cola Project (RAIN III). Implemented between February 2017 and May 2020, the RAIN III project aimed to expand the urban distribution network beyond the existing transmission system. The project included: (1) construction of four local service reservoirs and three booster pumping stations in newly served areas; (2) installation of 62 km of primary and secondary distribution pipelines serving northwestern and southern Hargeisa (shown in light blue areas in **Figure 6**), thereby extending piped water service to previously unconnected neighborhoods; and (3) construction of 32 water kiosks serving refugee and internally displaced persons (IDP) settlements.¹⁴

Hargeisa Water Supply & Basic Sanitation Project (KfW Project). Launched in February 2023 and scheduled for completion in late 2026, the KfW project has two sequential objectives: (1) expanding and upgrading neighborhood-level distribution infrastructure, and (2) extending household connections to the upgraded network.¹⁵

Phase I has been completed and included: (a) construction of four large reinforced-concrete service reservoirs located at elevated sites across different districts to provide gravity-fed distribution; (b) installation of 166 km of neighborhood-level distribution

¹³See [UN-Habitat \(2022\)](#) for detailed introduction to HUWSUP.

¹⁴See [Terre Solidali \(2017\)](#) and [Darood \(2025\)](#) for discussion on RAIN III.

¹⁵See [PPIAF and World Bank \(2018\)](#), [KfW Development Bank \(2019\)](#), and [Somaliland Standard \(2019\)](#) for coverage on KfW Project.

mains across six water supply zones; and (c) construction of 276 public water kiosks. Phase II is currently underway (as of May 2026) and focuses on connecting individual households to the newly installed distribution network. The service area of the KfW project is shown in medium blue color in **Figure 6**. Households in the service area can obtain access either through individual service contracts with HWA or through community-level agreements negotiated collectively with HWA.

4.2 Implications on water access

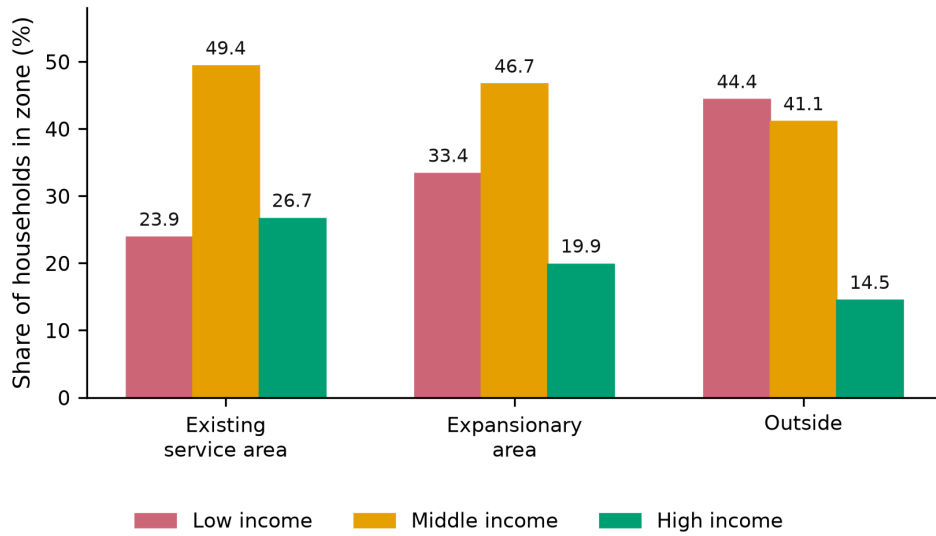
To understand the implications of the ongoing service expansion on households' water access and consumption, we compare household characteristics by service availability. Instead of grouping households by their current household-level connectivity, we categorize them by GPS location relative to piped water supply zones displayed in **Figure 6**: the existing service area (covered by the current network), the expansionary area (the KfW and Coca-Cola service zones, excluding the existing service area), and the outside area.¹⁶ Nearly all households currently connected to the network are located within the existing service area. Households in the expansionary area will have the opportunity to receive a connection, subject to negotiation outcomes. Households in the outside area will not receive a connection in the near future under current plans.

First, the ongoing network expansion is likely to mitigate the regressiveness of water access by income documented in Section 3.3. **Figure 9** shows that a higher share of low-income households (33.4%) reside in the expansionary area than in the existing service area (23.9%), which has skewed toward middle- and high-income households. Because the expansionary area draws from a more balanced cross-section of the income distribution, extending piped connections there would reduce the income gradient in access more effectively than the existing network configuration does.

Second, the expansion may ameliorate water shortage conditions. Households in the expansionary area currently experience higher frequencies of water shortage across all three dimensions — basic deprivation, activity disruption, and general worry — compared to those in the existing service area. Third, the overall cost of water is likely to decrease. Households in the expansionary area currently pay a blended unit price of \$7.14 per 1,000 liters, driven by dependence on tanker trucks and water carts. As these households gain access to piped water at approximately \$1.425 per cubic meter, their reliance on more expensive sources should diminish. (see Appendix **Table 6** for detailed comparison.)

¹⁶The KfW service area is based on the shapefile provided by HWA, which clearly defines the service boundary. In the absence of clearly defined boundaries for the other two networks, we extrapolate the existing network by setting a 30-meter buffer around the shapefile of household connection pipes in the current network. The Coca-Cola service area is extrapolated by matching KfW service density (km² of catchment per km of pipe).

Figure 9: Income Composition of Households by Supply Zone



Note: Values are population-weighted shares of self-reported monthly household income levels within each network supply zone. Low income is $\leq \$200$, middle income is $\$201-\500 , and high income is $\geq \$501$. Existing service area covers the inner city currently supplied by piped water ($N = 878$); expansionary area contains the KfW and Coca-Cola service zones, excluding the existing service area ($N = 574$); outside area is regions not covered in the expansion plan ($N = 745$).

However, potential gains from expansion are bounded. Households outside all service boundaries — who are disproportionately low-income (**Figure 9**), consume the least water, and experience the highest shortage rates (Appendix **Table 6**) — will not benefit from current infrastructure plans and will continue to face the most severe access constraints. Moreover, as the extensive margin of piped water access expands, service frequency to currently reliably connected households is scheduled to decrease due to limitations in total water extraction and transmission capacity. Connected households may consequently need to diversify their water portfolio toward costlier sources and invest more heavily in storage capacity.

5 Conclusion

This paper documents the structure of urban water insecurity in Hargeisa through a city-representative household survey and complementary qualitative fieldwork. We show that household water access is best understood not as a binary distinction between connected and unconnected households, but as a patchwork system in which families combine piped connections, tanker trucks, cart vendors, neighbor sharing, rainwater harvesting, and storage. These portfolios vary sharply across seasons and across households, with rainwater playing an especially important role during the rainy season.

Four findings stand out. First, most households rely on multiple sources, and this

source-stacking reflects the intermittency of nearly all available water options. Second, piped connections remain valuable despite unreliable service: households with more frequent network access pay lower unit prices, purchase more water, and experience fewer shortages. Third, the system is regressive. Low-income households are less likely to be connected, more likely to live outside the current network footprint, more exposed to shortages, and often pay more per liter for non-network water. Fourth, storage is a central coping technology, but one that is unevenly distributed. Poorer households have less capacity to store water precisely when storage would help them smooth unreliable supply, harvest rainwater, and reduce dependence on urgent or small-volume purchases.

The ongoing expansion of Hargeisa's piped network has the potential to improve water access and reduce some of these inequities, particularly for households in planned service areas that currently rely on expensive tanker and cart water. However, infrastructure expansion alone will not eliminate the patchwork system. Households outside planned service boundaries will continue to face severe constraints, and limits on total extraction and transmission capacity mean that new connections may come with reduced service frequency for some currently connected households. As a result, policies that expand piped access should be paired with attention to reliability, affordability, storage, rainwater harvesting, and the regulation of non-network providers. More broadly, the Hargeisa case shows why urban water security cannot be fully measured by primary source or connection status alone. Understanding how households combine sources, absorb costs, and cope with intermittency is essential for designing water systems that are both more resilient and more equitable.

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Appendix

A Population Weighted Tables

Table 1: Population Weighted: HH characteristics, storage, shortage, and water purchase by connectivity

Variable	HH-level Piped Water Connection Reliability			Overall p-value
	Unconnected	Low freq (≤ 3 day/week)	High freq (≥ 4 day/week)	
N (HHs)	1,545	466	186	
— Panel A. Household characteristics —				
Avg. household size	7.7	8.2	8.0	0.13
% Low income ($\leq$ \$200)	39.9%	19.7%	16.2%	< 0.01
% Middle income (\$201–\$500)	43.6%	53.1%	43.8%	0.07
% High income ($>$ \$500)	16.5%	27.1%	40.0%	< 0.01
— Panel B. Water storage capacity —				
N (with complete storage data)	1,073	305	108	
% With any storage	77.5%	86.5%	78.9%	0.30
% With total storage $\geq 1,000$ L	38.3%	31.2%	38.2%	0.52
Avg. total storage (L)	1,150.1	1,298.3	730.2	0.17
— Panel C. Reported water shortage —				
<i>C.1 Basic deprivation (drink, wash, food, clothing)</i>				
% Never	58.9%	81.0%	95.1%	< 0.01
% Occasional (once–twice in 4 weeks)	14.5%	11.1%	4.4%	0.03
% Weekly or more	26.5%	7.9%	0.5%	< 0.01
<i>C.2 Activity disruption (work, school)</i>				
% Never	79.2%	93.2%	99.1%	< 0.01
% Occasional (once–twice in 4 weeks)	8.9%	4.8%	0.9%	< 0.01
% Weekly or more	11.9%	2.0%	0.0%	< 0.01
<i>C.3 Worry about shortage for general HH needs</i>				
% Never	65.7%	84.9%	96.2%	< 0.01
% Occasional (once–twice in 4 weeks)	14.9%	8.2%	3.3%	< 0.01
% Weekly or more	19.3%	6.8%	0.5%	< 0.01
— Panel D. Water from carts (resurveyed cart users) —				
N (cart users, resurveyed)	49	19	0	
Avg. unit price (\$/1,000 L)	8.82	8.61	—	
Avg. weekly quantity (L)	1,512.2	2,441.7	—	
Avg. weekly spending (\$)	12.3	20.3	—	
— Panel E. Water from tanker trucks (resurveyed tanker users) —				
N (tanker users, resurveyed)	70	63	3	
Avg. unit price (\$/1,000 L)	7.06	7.09	7.00	0.02
Avg. weekly quantity (L)	1,012.8	983.5	2,258.2	< 0.01
Avg. weekly spending (\$)	7.2	6.9	15.8	< 0.01
— Panel F. Water from piped water —				
Assumed unit price (\$/1,000 L)	1.425	1.425	1.425	
Avg. weekly quantity (L)	0.0	797.9	3,313.0	< 0.01
Avg. weekly spending (\$)	0.0	1.1	4.7	< 0.01
— Panel G. Total water purchase —				
N (with valid water purchase data)	109	79	185	
Avg. weekly total quantity (L)	1,047.8	2,028.7	3,382.1	< 0.01
Avg. weekly per-person quantity (L)	144.4	239.5	486.9	< 0.01
Avg. weekly total spending (\$)	7.5	10.3	5.2	< 0.01
Avg. overall unit price (\$/1,000 L)	7.15	5.81	1.51	< 0.01

Notes — All means, shares, and p -values are population-weighted. Weights are inverse zone-level inclusion probabilities calibrated to known stratum population sizes. Resurvey-based outcomes carry augmented weights adjusting for second-phase selection and completion probabilities. Reported N values are unweighted observation counts. Households are grouped by their piped network connection and service frequency. Storage statistics (Panel B) pool rainwater collectors with main-survey storage data and non-rainwater users resurveyed to fill in storage measurements skipped in the main survey, using storage-specific augmented weights. Storage means are unconditional on positive storage. Cart and tanker panels (D, E) use resurvey-specific augmented weights and are restricted to observed users of each source. Estimates for the *High-freq* group in Panel D are suppressed (shown as —) because the main survey identifies no cart users in this group. Water from piped water (Panel F) uses a hybrid sample selection: for *High-freq* group, all households in the main survey are included; for *Unconnected* and *Low-freq* groups, we include resurveyed households only and report their utility spending from the main survey. Utility quantity is derived from dividing utility spending by the HWA piped tariff of \$1.425/1,000L. Total water purchase (Panel G) follows the same sample selection as Panel F, and sums cart, tanker, and utility components; non-purchase from any source is coded as zero. p -values are design-adjusted Wald tests of the joint null that all connectivity-group dummy coefficients equal zero in a survey-weighted linear regression, with Taylor-linearized standard errors clustered by zone and stratified by the original sampling strata.

Table 2: Population Weighted: Piped access, storage, shortage, and water purchase by household income

Variable	Reported Monthly Income Level			Overall p-value	Low vs High p-value
	Low ($\leq$ \$200)	Middle (\$201-500)	High ($\geq$ \$501)		
N (HHs)	789	1,009	393		
— Panel A. HH-level piped network connection —					
% Unconnected	85.5%	70.5%	59.9%	< 0.01	< 0.01
% Low-freq connection (0–3 days/wk)	11.1%	22.6%	26.0%	< 0.01	< 0.01
% High-freq connection (4–7 days/wk)	3.4%	6.9%	14.1%	0.03	0.01
— Panel B. Piped water supply zone —					
% Existing service area	24.7%	38.5%	46.8%	< 0.01	< 0.01
% Expansionary area	25.0%	26.3%	25.2%	0.90	0.97
% Outside	50.3%	35.1%	28.0%	< 0.01	< 0.01
— Panel C. Reported water shortage (general HH need) —					
% Never	58.0%	75.7%	85.4%	< 0.01	< 0.01
% Occasional (once–twice in 4 weeks)	18.7%	10.3%	8.4%	< 0.01	< 0.01
% Weekly or more	23.1%	14.0%	6.2%	< 0.01	< 0.01
— Panel D. Water storage capacity —					
N (HHs with full storage data)	490	713	279		
% With any storage	71.2%	86.9%	76.2%	0.29	0.24
% With total storage $\geq$ 1,000L	30.5%	42.4%	35.5%	0.42	0.35
Avg. total storage (L)	814.1	1,180.8	1,632.6	0.06	0.02
— Panel E. Water purchase from carts (resurveyed cart users) —					
N (cart users, resurveyed)	28	32	8		
Avg. unit price (\$/1,000 L)	9.14	8.47	8.04	0.02	< 0.01
Avg. weekly quantity (L)	1,359.6	1,497.6	4,709.8	0.22	0.09
Avg. weekly spending (\$)	11.7	12.3	34.8	0.22	0.08
— Panel F. Water purchase from tanker trucks (resurveyed tanker users) —					
N (tanker users, resurveyed)	46	46	44		
Avg. unit price (\$/1,000 L)	6.96	7.20	6.96	0.02	0.97
Avg. weekly quantity (L)	853.7	962.4	1,451.3	< 0.01	< 0.01
Avg. weekly spending (\$)	5.9	7.0	10.2	< 0.01	< 0.01
— Panel G. Water purchase from piped utility (resurveyed utility payers) —					
Assumed unit price (\$/1,000 L)	1.42	1.42	1.42		
Avg. weekly quantity (L)	34.1	94.4	106.8	< 0.01	0.03
Avg. weekly spending (\$)	0.0	0.1	0.2	< 0.01	0.03
— Panel H. Total water purchase (all resurveyed HHs) —					
N (resurveyed HHs)	69	74	48		
Avg. weekly total quantity (L)	941.7	1,087.8	1,687.1	< 0.01	< 0.01
Avg. weekly per-person quantity (L)	144.0	133.5	221.4	< 0.01	< 0.01
Avg. weekly total spending (\$)	6.5	7.4	11.3	< 0.01	< 0.01
Avg. overall unit price (\$/1,000 L)	7.04	7.10	6.80	0.02	0.01

Notes — All means, shares, and p -values are population-weighted. Weights are computed as inverse zone-level inclusion probabilities calibrated to known stratum population sizes (main-survey outcomes); for resurvey-based outcomes, main-survey weights are additionally augmented for second-phase selection and completion probabilities and rescaled within income-by-connection cells. Reported N values are unweighted observation counts. Households are grouped by self-reported monthly income: Low ($\leq$ \$200), Middle (\$201–\$500), High ($>$ \$500); households reporting “Don’t know” or “Prefer not to answer” are excluded. Panel C pools rainwater-user (RW) households (main-survey storage) with resurveyed non-rainwater-user (non-RW) households; non-RW HHs not resurveyed are excluded. Cart and tanker panels use resurvey-based weights and are restricted to observed users of each source. Utility panel assumes a flat piped tariff of \$1.425/m³. Total water purchase sums cart, tanker, and utility components, with non-purchase from any source coded as zero. Overall p -values are design-adjusted Wald tests of the joint null that all income-group dummy coefficients equal zero in a survey-weighted linear regression of the row outcome on income-group indicators, with Taylor-linearized standard errors clustered by zone and stratified by the original sampling strata. Low vs high p -values are design-adjusted Wald tests of the null that high-income dummy coefficient equal zero in the linear regression of the row outcome on income-group indicators.

Table 3: Population Weighted: HH characteristics, storage, shortage, and water purchase by supply zone

Variable	Network Service Area			Overall p-value
	Existing	Expansionary (KfW + Coca-Cola)	Outside	
N (HHs in zone)	878	574	745	
— Panel A. Household characteristics —				
Avg. household size	8.0	7.9	7.7	0.39
N (with income reported)	875	572	744	
% Low income ($\leq$ \$200)	23.9%	33.4%	44.4%	< 0.01
% Middle income (\$201–\$500)	49.4%	46.7%	41.1%	0.23
% High income ($>$ \$500)	26.7%	19.9%	14.5%	0.02
— Panel B. Water storage capacity —				
N (HHs with full storage data)	569	412	505	
% With any storage	83.9%	69.8%	80.3%	0.26
% With total storage $\geq$ 1,000L	29.9%	26.7%	51.2%	< 0.01
Avg. total storage (L)	870.4	1,128.0	1,475.1	0.16
— Panel C. Reported water shortage —				
<i>C.1 Basic deprivation (drink, wash, food, clothing)</i>				
% Never	80.6%	56.8%	58.2%	< 0.01
% Occasional (once–twice in 4 weeks)	10.6%	17.6%	12.6%	0.08
% Weekly or more	8.8%	25.6%	29.3%	< 0.01
<i>C.2 Activity disruption (work, school)</i>				
% Never	91.4%	82.3%	76.6%	< 0.01
% Occasional (once–twice in 4 weeks)	5.0%	8.7%	9.1%	0.08
% Weekly or more	3.6%	9.1%	14.3%	< 0.01
<i>C.3 Worry about shortage for general HH need</i>				
% Never	84.6%	65.1%	64.1%	0.01
% Occasional (once–twice in 4 weeks)	8.8%	14.1%	15.6%	0.18
% Weekly or more	6.6%	20.8%	20.3%	< 0.01
— Panel D. Water from carts (resurveyed cart users) —				
N (cart users, resurveyed)	46	21	1	
Avg. unit price (\$/1,000 L)	8.75	8.84	9.00	0.43
Avg. weekly quantity (L)	1,695.2	1,656.9	3,000.0	0.07
Avg. weekly spending (\$)	14.0	13.2	27.0	0.05
— Panel E. Water from tanker trucks (resurveyed tanker users) —				
N (tanker users, resurveyed)	67	41	28	
Avg. unit price (\$/1,000 L)	7.15	7.05	7.03	0.31
Avg. weekly quantity (L)	970.9	1,112.1	952.7	0.27
Avg. weekly spending (\$)	6.9	8.0	6.7	0.15
— Panel F. Water from piped water (resurveyed HHs) —				
N (utility payers, resurveyed)	105	57	29	
Assumed unit price (\$/1,000 L)	1.42	1.42	1.42	
Avg. weekly quantity (L)	292.5	3.4	0.0	0.04
Avg. weekly spending (\$)	0.4	0.0	0.0	0.04
— Panel G. Total water purchase (resurveyed HHs) —				
N (resurveyed HHs)	105	57	29	
Avg. weekly total quantity (L)	1,396.1	1,166.1	956.4	0.05
Avg. weekly total spending (\$)	8.6	8.4	6.7	0.05
Avg. overall unit price (\$/1,000 L)	6.85	7.14	7.03	0.37

Notes — All means, shares, and p -values are population-weighted. Weights are inverse zone-level inclusion probabilities calibrated to known stratum population sizes. Resurvey-based outcomes carry augmented weights adjusting for second-phase selection and completion probabilities. Reported N values are unweighted observation counts. Households are grouped by GPS location relative to piped water supply zones: Existing network, Expansionary area, and outside area. p -values are design-adjusted Wald tests of the joint null that all supply-zone dummy coefficients equal zero, with Taylor-linearized standard errors clustered by zone and stratified by the original sampling strata.

B Unweighted Sample Tables

Table 4: Unweighted: HH characteristics, storage, shortage, and water purchase by connectivity

Variable	HH-level Piped Water Connection Reliability			p-value
	Unconnected	Low freq (≤ 3 day/week)	High freq (≥ 4 day/week)	
N (HHs)	1,545	466	186	
— Panel A. Household characteristics —				
Avg. household size	7.7	8.0	7.9	0.10
N (with income reported)	1,541	465	185	
% Low income ($\leq$ \$200)	42.5%	20.6%	20.5%	< 0.01
% Middle income (\$201–\$500)	43.5%	54.2%	47.0%	< 0.01
% High income ($>$ \$500)	14.0%	25.2%	32.4%	< 0.01
— Panel B. Water storage capacity —				
N (HHs with full storage data)	1,073	305	108	
% With any storage	85.7%	90.5%	84.3%	0.07
% With total storage $\geq 1,000$ L	35.6%	28.2%	36.1%	0.05
Avg. total storage (L)	1,320.2	1,255.4	934.9	0.22
— Panel C. Reported water shortage —				
<i>C.1 Basic deprivation (drink, wash, food, clothing)</i>				
% Never	57.1%	83.7%	93.5%	< 0.01
% Occasional (once–twice in 4 weeks)	15.3%	9.7%	5.9%	< 0.01
% Weekly or more	27.6%	6.7%	0.5%	< 0.01
<i>C.2 Activity disruption (work, school)</i>				
% Never	78.5%	94.0%	98.9%	< 0.01
% Occasional (once–twice in 4 weeks)	10.0%	4.3%	1.1%	< 0.01
% Weekly or more	11.4%	1.7%	0.0%	< 0.01
<i>C.3 Worry about shortage for general HH needs</i>				
% Never	63.8%	89.1%	95.7%	< 0.01
% Occasional (once–twice in 4 weeks)	16.6%	6.4%	3.8%	< 0.01
% Weekly or more	19.6%	4.5%	0.5%	< 0.01
— Panel D. Water from carts (resurveyed cart users) —				
N (cart users, resurveyed)	49	19	0	
Avg. unit price (\$/1,000 L)	8.67	8.53	—	
Avg. weekly quantity (L)	1,553.1	2,078.9	—	
Avg. weekly spending (\$)	12.5	17.3	—	
— Panel E. Water from tanker trucks (resurveyed tanker users) —				
N (tanker users, resurveyed)	70	63	3	
Avg. unit price (\$/1,000 L)	6.95	7.04	7.00	
Avg. weekly quantity (L)	1,057.1	987.3	2,000.0	0.59
Avg. weekly spending (\$)	7.4	6.9	14.0	0.57
— Panel F. Water from piped water —				
N (utility payers)	—	79	185	
Assumed unit price (\$/1,000 L)	—	1.43	1.43	
Avg. weekly quantity (L)	0.0	847.2	3,250.5	
Avg. weekly spending (\$)	0.0	1.2	4.6	
— Panel G. Total water purchase —				
N	109	79	185	
Avg. weekly total quantity (L)	1,377.1	2,134.6	3,282.9	< 0.01
Avg. weekly per-person quantity (L)	195.5	249.6	478.4	< 0.01
Avg. weekly total spending (\$)	10.4	10.9	4.9	< 0.01
Avg. overall unit price (\$/1,000 L)	7.71	5.70	1.46	< 0.01

Notes — The table reports unweighted values from the sample, covering identical measures as Table 1.

Table 5: Unweighted: Piped access, storage, shortage, and water purchase by household income

Variable	Reported Monthly Income Level			p-value
	Low ($\leq$ \$200)	Middle (\$201-500)	High ($\geq$ \$501)	
N (HHs)	789	1,009	393	
— Panel A. HH-level piped network connection —				
% Unconnected	83.0%	66.4%	55.0%	< 0.01
% Low-freq connection (0–3 days/wk)	12.2%	25.0%	29.8%	< 0.01
% High-freq connection (4–7 days/wk)	4.8%	8.6%	15.3%	< 0.01
— Panel B. Piped water supply zone —				
% Existing network	29.0%	43.5%	52.7%	< 0.01
% Expansion (servable)	27.2%	26.5%	22.9%	0.26
% Outside	43.7%	30.0%	24.4%	< 0.01
— Panel C. Water storage capacity —				
N (HHs with full storage data)	490	713	279	
% With any storage	80.4%	89.2%	90.7%	< 0.01
% With total storage $\geq$ 1,000L	32.7%	33.6%	38.0%	0.30
Avg. total storage (L)	1,008.0	1,146.8	2,061.6	< 0.01
— Panel D. Reported water shortage (general HH need) —				
% Never	59.2%	76.5%	85.5%	< 0.01
% Occasional (once–twice in 4 weeks)	19.5%	10.4%	8.4%	< 0.01
% Weekly or more	21.3%	13.1%	6.1%	< 0.01
— Panel E. Water purchase from carts (resurveyed cart users) —				
N (cart users, resurveyed)	28	32	8	
Avg. unit price (\$/1,000 L)	9.02	8.44	8.06	0.13
Avg. weekly quantity (L)	1,496.4	1,440.6	3,450.0	0.53
Avg. weekly spending (\$)	12.7	11.8	26.0	0.51
— Panel F. Water purchase from tanker trucks (resurveyed tanker users) —				
N (tanker users, resurveyed)	46	46	44	
Avg. unit price (\$/1,000 L)	6.94	7.13	6.90	0.02
Avg. weekly quantity (L)	773.9	962.0	1,417.0	< 0.01
Avg. weekly spending (\$)	5.4	6.9	9.8	< 0.01
— Panel G. Water purchase from piped utility (resurveyed utility payers) —				
N (utility payers, resurveyed)	69	74	48	
Assumed unit price (\$/1,000 L)	1.42	1.42	1.42	
Avg. weekly quantity (L)	272.6	502.6	377.6	0.28
Avg. weekly spending (\$)	0.4	0.7	0.5	0.28
— Panel H. Total water purchase (all resurveyed HHs) —				
N (resurveyed HHs)	69	74	48	
Avg. weekly total quantity (L)	1,395.8	1,723.6	2,251.5	0.10
Avg. weekly per-person quantity (L)	192.7	223.8	259.8	0.25
Avg. weekly total spending (\$)	9.1	10.1	13.8	0.27
Avg. overall unit price (\$/1,000 L)	7.14	6.80	6.39	0.11

Notes — The table reports unweighted values from the sample, covering identical measures as Table 2.

Table 6: Unweighted: HH characteristics, storage, shortage, and water purchase by supply zone

Variable	Network Service Area			Overall p-value
	Existing	Expansionary (KfW + Coca-Cola)	Outside	
N (HHs in zone)	878	574	745	
— Panel A. Household characteristics —				
Avg. household size	7.9	7.7	7.7	0.34
N (with income reported)	875	572	744	
% Low income ($\leq$ \$200)	26.2%	37.6%	46.4%	< 0.01
% Middle income (\$201–\$500)	50.2%	46.7%	40.7%	< 0.01
% High income ($>$ \$500)	23.7%	15.7%	12.9%	< 0.01
— Panel B. Water storage capacity —				
N (HHs with full storage data)	569	412	505	
% With any storage	90.0%	83.5%	85.3%	< 0.01
% With total storage $\geq$ 1,000L	28.1%	30.7%	43.8%	< 0.01
Avg. total storage (L)	995.5	1,269.0	1,606.2	< 0.01
— Panel C. Reported water shortage —				
<i>C.1 Basic deprivation (drink, wash, food, clothing)</i>				
% Never	81.6%	57.3%	53.8%	< 0.01
% Occasional (once–twice in 4 weeks)	9.9%	15.2%	15.9%	< 0.01
% Weekly or more	8.4%	27.4%	30.3%	< 0.01
<i>C.2 Activity disruption (work, school)</i>				
% Never	91.8%	83.4%	74.0%	< 0.01
% Occasional (once–twice in 4 weeks)	5.0%	8.0%	11.6%	< 0.01
% Weekly or more	3.2%	8.6%	14.4%	< 0.01
<i>C.3 Worry about shortage for general HH need</i>				
% Never	86.4%	64.4%	60.6%	< 0.01
% Occasional (once–twice in 4 weeks)	8.2%	14.5%	18.4%	< 0.01
% Weekly or more	5.4%	21.1%	21.0%	< 0.01
— Panel D. Water from carts (resurveyed cart users) —				
N (cart users, resurveyed)	46	21	1	
Avg. unit price (\$/1,000 L)	8.60	8.69	9.00	
Avg. weekly quantity (L)	1,708.7	1,619.0	3,000.0	
Avg. weekly spending (\$)	14.0	12.9	27.0	
— Panel E. Water from tanker trucks (resurveyed tanker users) —				
N (tanker users, resurveyed)	67	41	28	
Avg. unit price (\$/1,000 L)	7.01	6.97	6.98	0.86
Avg. weekly quantity (L)	989.6	1,145.1	1,033.9	0.57
Avg. weekly spending (\$)	6.9	8.0	7.2	0.55
— Panel F. Water from piped water (resurveyed HHs) —				
N (utility payers, resurveyed)	105	57	29	
Assumed unit price (\$/1,000 L)	1.42	1.42	1.42	
Avg. weekly quantity (L)	692.6	24.6	0.0	
Avg. weekly spending (\$)	1.0	0.0	0.0	
— Panel G. Total water purchase (resurveyed HHs) —				
N (resurveyed HHs)	105	57	29	
Avg. weekly total quantity (L)	2,072.6	1,444.8	1,101.7	< 0.01
Avg. weekly per-person quantity (L)	269.8	163.4	161.5	< 0.01
Avg. weekly total spending (\$)	11.5	10.6	7.9	0.02
Avg. overall unit price (\$/1,000 L)	6.35	7.56	7.05	< 0.01

Notes — The table reports unweighted values from the sample, covering identical measures as Table 6.